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Enhancing Optimized Personalized Therapy in Clinical Decision Support System using Natural Language Processing

Basavaraj N. Hiremath^{a,*}, Malini M. Patil^{a,b}^a Department of Computer Science and Engineering, J S S Academy of Technical Education, Bengaluru-560060, Karnataka, India^b Department of Information Science and Engineering, J S S Academy of Technical Education, Bengaluru-560060, Visvesvaraya Technological University, Belagavi-590018, Karnataka, India

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ABSTRACT

Sentiment analysis is the process of identifying and categorising the opinions expressed by human utterances through computational techniques using natural language processing. The present work focuses on a case study to develop a clinical decision support system for personalized therapy process using aspect-based sentiment analysis. The process is carried out on a drug review data in order to determine whether the patient's behaviour towards a medicine, product, treatment etc is positive, negative or neutral using NLP techniques. The polarities obtained are compared for further analysis of the patient reviews for the better clinical decision system. Machine learning methods are also used for classification of the drug review data to compare the sentiment scores. The prominent statistical sklearn models used are support vector machines (SVM), Random Forest Classification, LinearSVC, MultinomialNB. SVM algorithm is found to perform better compared to other in terms of accuracy.

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1. Introduction

Human linguistic communication occurs through speech. Most of the activities in communication happen through speech. Growth of technology laid the foundation of man machine interface. The ability of a machine to understand human linguistics is referred to as natural language processing (NLP). In the context of NLP, the authors (Cambria et al., 2017b) points to big suitcase comprising all analysis like polarity, subjectivity which categorises into pragmatics, semantics and syntactic layers. The entire language processing problem can be classified into two major tasks. One is processing by lexical, syntactic and semantic knowledge of the language on written text and other is processing by spoken language using the phonology. NLP is a process which necessitates use of algorithms to bring out and identify rules based in natural language. The important applications of NLP are Google Translator, Words Grammarly Check, IVR Systems, Google Personal Assistants,

Siri, Alexa, Cortana, Ok Google. NLP plays a significant role in assisting machine-human interactivity.

The conversation between human to human is by way of written or oral communication. Understanding the syntax and semantics of the language is the inborn quality of a human being. To accomplish the effective talk and response to complete the communication (Hiremath and Patil, 2019a) it is imperative to understand the language constructs. This requires frequent and voluminous interactions between machines and humans. The mode of interaction can be termed as a data (Hiremath and Patil, 2019b). This unstructured language data is transformed into code which the computer can understand. When the text data is processed as input, the computer runs algorithms to bring out meanings correlated with every sentence and collect the significant information. The process cycle for transfer of information or exchange of data between human and machine happens in various forms. The entire flow is drawn in the form of a chart as shown in Fig. 1.

Sentiment analysis and Affective computing (Cambria et al., 2017a) have shown potential in analysing human communication in artificial intelligence. The authors have developed a hybrid framework for polarity detection, majorly on emotion recognition tasks. Affective computing results in categorising emotion labels, which are mainly derived out of shared social expressions. The authors (Cambria et al., 2012), have explained about sentiment

* Corresponding author.

E-mail address: basavaraj@ieee.org (B.N. Hiremath).

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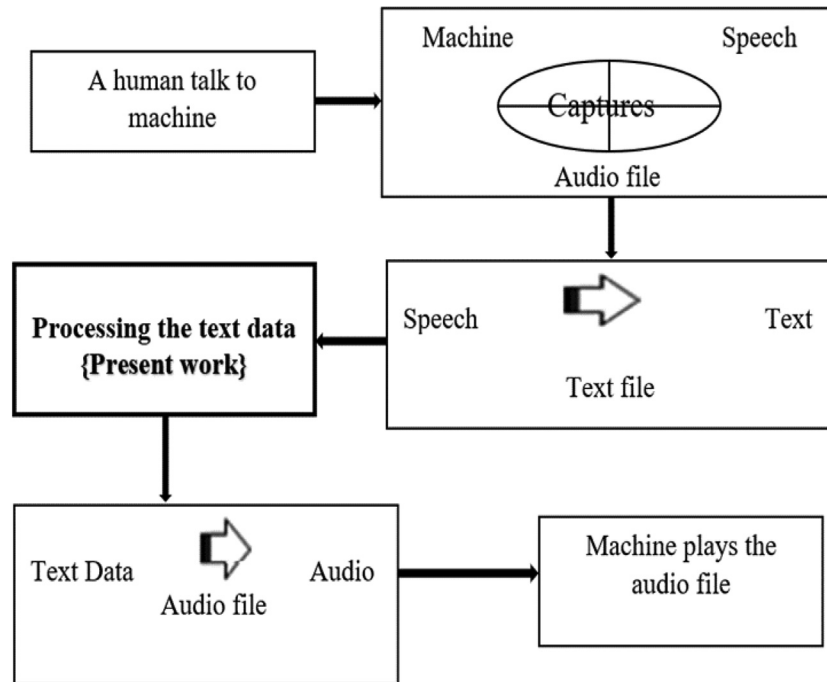


Fig. 1. Interface between human and machine.

analysis of standard patient reported outcome measures, as a measure to monitor outcomes from patient's data. The data will be in free text expressed in their satisfactory feelings or demonstrative complaining about physio emotional experiences of health care. These sentic computing methods are categorised as sentic dimensions like sensitivity, attention, pleasantness and attitude on frustration, anxiety, optimism, rejection and disapproval. The work (Gräßer et al., 2018) deals with aspect-based opinion analysis with transfer learning methodology. It emphasises on the analysis of pharmaceutical reviews, model transferability with in-domain i.e. condition-based training information which enrich the evaluation with high performance classification results. It concludes the findings of support to pharmacovigilance and improvement of clinical decision support system.

The objective of the present work is to understand the different approaches to perform sentiment analysis and know the sentiment of persons by evaluating series of words on sentences of human language. The selected data is from reviews expressed by patient's experience on use of particular drugs for the specific therapy. In the health care domain, the post-marketing drug surveillance study, i.e. pharmacovigilance, plays a major role in evaluating drug safety, once a drug has been released. The clinical decision support systems are used to enhance optimized personalized therapy. The review sentences include identification of effectiveness of a drug, and its side effects. These two benefits are used to classify in aspect-based sentiment analysis.

The remaining part of this paper is outlined as follows: Section 2 is about preliminaries followed with literature survey in Section 3. The fourth section is about methods and models. Experiments, discussion and results are explained in Section 5. The paper concludes with conclusion and future work in the last section.

2. Preliminaries

NLP is the subfield of artificial intelligence that deals with computational algorithms which supports computers and humans' interactions.

(i) *Objective of NLP*: The objective of NLP is to perceive, analyse, read and decipher to interpret the human languages in a manner that is intelligible. Most of the NLP techniques rely on machine learning to derive meaning from human languages. It is also the most common text classification technique that analyses an incoming message and tells whether the sentiment of the incoming message is labelled as positive, negative or neutral.

(ii) *Challenges in NLP*: The task of teaching machines to understand the nature of human communication language is very difficult. Some tasks are at higher levels and others at detailed level. Comprehending the human language requires understanding at the word level and the concept level, how both are connected to deliver the intended message. The human's cognitive ability can easily master a language, whereas the obscurity and loosely formed characteristics of the natural language make NLP arduous for machines to implement.

(iii) *How NLP works?* The five steps of NLP are explained in brief with their specific functionalities as depicted in Fig. 2.

1. *Morphological analysis*: Individual words are analysed into their components. Separation of non-word tokens. Assigning syntactic categories to all the words.
2. *Syntactic analysis*: Relationship among the individual words is processed. Linear.

Sequence of words is checked. Rejection of sentences which do not follow the specified language rules.

3. *Semantic analysis*: Refers to the correct meaning of the sentences.
4. *Discourse integration*: The meaning of an individual sentence may depend on the proceeding context.
5. *Pragmatic analysis*: Deals with interpretation of sentences.

3. Literature survey

Sentiment analysis is a nascent research, over the years the associated challenges has increased (Valdivia et al., 2017) like

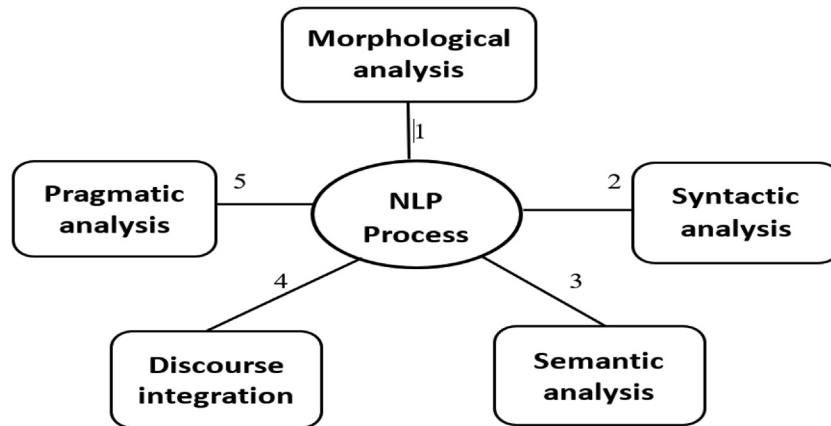


Fig. 2. NLP process cycle.

opinion summarization, retrieval and subjectivity classification. This has enabled validation process by statistical measures (Wilson and Hernández-Hall, 2014). The authors (Laksono et al., 2019) have concluded that statistical classification like Naïve Bayes method for identifying sentiment analysis gave better results than Textblob, a python based open source library used to carry out all the NLP functions, and to measure quantitative assessment of sentiment in a sentence.

In (Hiremath and Patil, 2017) the authors describe a computer system for understanding English with respect to NLP. The computer system basically uses an interactive method for English language. The system answers questions, executes commands, and accepts information. The basic aspects of modelling language namely syntax, semantics and inference are considered. The system has programs for semantic analysis and consists of a parser, a recognition grammar, and a general solution framework system. In another work (Fernández-Gavilanes et al., 2016) sentiment analysis approach is discussed to extract sentiments. The paper also focuses on the expression of sentiments by classifying the essential issues as positive (favourable) or negative (unfavourable) opinions toward the subject. The authors in (Daelemans and Hoste, 2002) framed a solution to the problem where predicting a word from preceeding words in a sample sentence of text using n-gram models. Authors also proposed several statistical algorithms for assigning words to groups based on the frequency of their congruence with other words. Some of the works found in (Brown et al., 1992) focuses on the use of text analysis on human genome which is one of the breakthroughs works found. Also, in (Turian and Ph, 2013) it is found that text analytics is considered as one of the important works found in enterprise data initiatives.

The NLP is an important process in linguistic analysis of any human language. The researchers (Vyas and Uma, 2018) have made a review on various tools using NLP methods to identify sentiment analysis of texts. The subjectivity classification of texts was done in experiments of sentiment analysis on Arabic language texts by the authors (Boudad et al., 2017), where they have described about lexicon-based approaches. The polarity expansion methods are used (Fernández-Gavilanes et al., 2016) with semi-automatic techniques as a novel method to predict sentiment in review-based datasets, as it is used in this work. In paper (Bharti et al., 2015) authors used TextBlob to carryout parts of speech tagging for the sentiment recognition of text in social media data. The NLP tools are used to perform tweet categorization, location extraction and traffic level extraction with methods like tokenization, sentence segmentation and basic tasks. A research (Sarkar et al., 2019) on credibility of sentiment is observed by computing

Textblob sentiment analyser and further the method calculates the fake observation of the publication on the internet.

Aspect term is a feature in sentiment classification based on the location and content of opinion words which are dealt in research (Park et al., 2020), a method, in which bidirectional long short-term memory is used. The qualitative analysis deals with content attention for example, “great food but the service was dreadful”. The aspect ‘food’ gets positive whereas the word ‘service’ gets labelled as negative, because of the location of word ‘great’ and ‘dreadful’ in the sentence. Deep learning model classification achieve comparable results. The authors (Cambria et al., 2017b) describes use of deep learning architecture in NLP by comparing issues like dependency, transparency and consistency using machine learning frameworks.

In the tfidf vectorizer which generates weighted unigrams, the authors (He and Zheng, 2019) have pointed that the Textblob and Vader are general purpose sentiment analysers which create weighted unigrams, in which the word library may perform better for general purpose sentiment analysis but may not be good performing packages when we use them for context specific dataset. Naïve Bayes classifier presents a better performance than Textblob analyser in the analysis of customer related survey data to choose restaurant by the authors (Laksono et al., 2019). Authors (Daelemans and Hoste, 2002) have provided information about unreliable interaction between the parameters of algorithm settings and selection of features within a single algorithm. The authors (Chaturvedi et al., 2018) state that subjectivity identification is crucial in increasing the accuracy in opinion mining, as this alone task in NLP which removes ‘factual’ content. The text with features based on similarity embedding words are used in carrying sentiment analysis, NLP techniques can be extended to extract features of prosodic, lexical, pragmatic and syntactic. The research (Suhaimin et al., 2017) points to usage of nonlinear support vector machine to classify texts.

4. Methods and models

This section focuses on the details of data set description, tools and methods for sentiment labelling, sentiment scores.

4.1. Data set description

The archive data set of UCI University of California Irene specifically made available for analysing empirical computation through machine learning algorithms using available computation libraries.

The data set selected in this repository belongs to drug review data set. The detailed data description is displayed in Table 1. Number of instances are 215 thousand and number of attributes are six, which comprises of sentences in English language. These depicts about feedback and summaries of the drug experience. Learning split is 75% for training set and 25% for testing set. Useful count is the attribute which holds value of number of users who reviewed and shared.

4.2. NLP tools

The Vader (Valence Aware Dictionary for Sentiment Reasoning), is a tool for sentiment analysis to find good predictive accuracy to measure emotional state. Vader lexicon was originally developed by C.J hutto (Sarkar, 2019) based on rule-based methods, The Vader doesn't need to be trained (Tomp et al., 2019) as it is built on a lexicon with standard sentiment library. The sentiment lexicon used by Vader is validated as gold standard, (Sarkar, 2019) tested by humans, whereas the Textblob is a python string. The Textblob is exclusively used by researchers (Mishra et al., 2019) for analysing context-based sentiment analysis of social media dataset with image as context source. Textblob is a library used in python programming language for doing common text processing operations which is built on NLTK and patterns (Loria, 2015).

Table 1
Dataset description.

Sl no	Attribute	Datatype	Description
1	id	Integer	Unique id
2	drugname	Categorical	Name of drug
3	condition	Categorical	Name of condition
4	review	text	Patient review
5	Rating	numerical	10star rating
6	date	yyyy-mm-dd	Date of entry of review
7	useful count	numerical	Count of users who found useful about drug

4.3. Classifiers

For classification of patterns of data, the prominent statistical sklearn models are used i.e. Support Vector Classifier (Sarkar et al., 2019), Multinomial Naïve Bayes (Griol et al., 2017), Logistic Regression (Sarkar et al., 2019), Random Forest Classifier (Karoui et al., 2017) are used for the purpose of experiment and validated for 5 cross validation.

4.4. Sentiment labelling in NLP

Sentiment is an attitude or opinion that is influenced by emotion. Sentiment analysis is the process of classifying the polarity of a given text. Sentiment labelling is one way of classifying the sentences with negative or positive meaning. These sentences are derived from customer reviews, movie reviews or political remarks etc. The researchers (Chen et al., 2017) have categorically fine-tuned the reviews as very positive, neutral or very negative. The fundamental base to label is by way of manual labelling, in which degree of numbering is done. The machines can do the auto labelling with a standardized labelling score called as polarity score. It is built on sentiment library developed by each package. The labels will be used to classify and then predict the respective set by the use of learning algorithms.

4.5. Sentiment score in NLP

The sentiment score is the basic numbering to be summarised at a sentence level to arrive at required classification as negative, neutral or positive meaning. In programming languages lexicons called as sentiment polarity libraries are used. For example (Vechtomova, 2017) AFINN has 2477 valence labels, ANEW has 1034 words, MPQA has 8222 words. These are labelled as sentiment strengths (weak and strong) with prior polarity scores. But the authors (Rosenthal et al., 2014) have stressed the need of aspect based sentiment analysis, as a method to evaluate the polarity on the basis of key word called as "aspect". The polarity score is measured with a named tuple called polarity and subjectivity. The

Methodology for Sentiment labelling.

Input: dataset: = Drug reviews repository (UCI)

Result: Sentiment labels [as positive, negative], **Classification:** = Validation by accuracy scores, summary of categories with high count of polarity by disease

Step 1: Input the Data set

Data source is in the tab separated value format of table with seven attributes, a method of data frame is used in 'pandas' package to handle the table structure as frames.

Step 2: Metadata validation: Metadata of table form is validated as per data dictionary.

Step 3: Pre-processing: Pre-processing of text data is done by using NLTK modules which includes punkt libraries for stop words. The intermediate results are stored as .csv file for further processing of tasks of classification.

Step 4: Installations: All the necessary required package libraries compatible with python programme should be installed like TextBlob, NLTK, Scikit.

Step 5: Computation: The rules are computed for calculating polarity score of the sentences. For feature vectorization the library of tfidf is used for extraction. The text blob uses the model for classification which uses support vector classifier with rbf kernel

Step 6: Result evaluation: It is tested with confusion matrix, to do the evaluation of classification. Statistical accuracy of the method is calculated for comparison analysis.

Fig. 3. Methodology for sentiment labelling.

polarity score ranges from -1 to 1 , the subjectivity from 0.0 to 1.0 . This process is widely used in Textblob. But in Vader analysis it is termed as positive, positive and negative, when it is $\geq (0.05)$, $\leq (-0.05)$ and rest, respectively.

4.6. Methodology for sentiment labelling

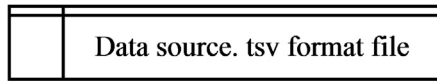
The steps followed in sentiment labelling are shown in Fig. 3 and are displayed as a flowchart in Fig. 4. The method focusses

on to classify and compute sentiment scores (negative sentiment, positive sentiment labels).

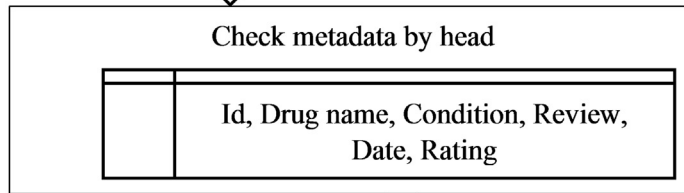
5. Experiments, results and discussions

This section focuses on the detailed discussion on the set of experiments conducted on sentiment labelling using Textblob and Vader packages based on the disease category of the dataset and prediction of sentiment labels is presented in the classification report.

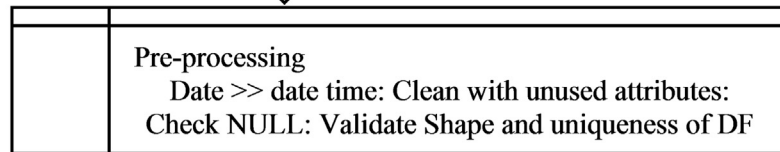
Step 1:



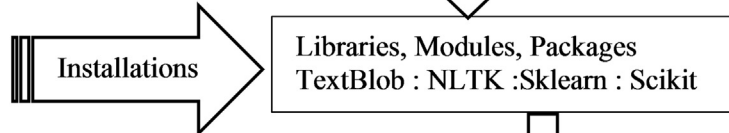
Step 2:



Step 3:

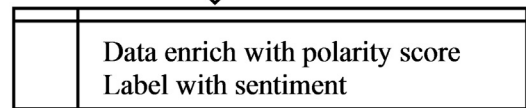


Step 4:



Step 5:

Sentiment Analyser



Step 6:

Review Score

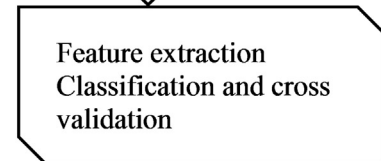


Fig. 4. Flow of algorithm.

Table 2

Aspect based tokenization for sample sentence.

Sl No	Sentence	Polarity	Subjectivity	Sentiment Label
1	"Suffered depression symptoms, as I continued medication depression got worse. Stopped taking it and it took 1 week for depression to diminish."	0.6	-0.4	Negative
2	"I have been taking Beyaz for exactly a month now and my acne has cleared up so much! I have always had really bad skin and It has made it flawless! It is amazing! I love Beyaz!"	0.602778	0.3625	Positive
3	"Longer acting steroid option. Gained weight and anxiety."——Addison's Disease	0	0	Negative

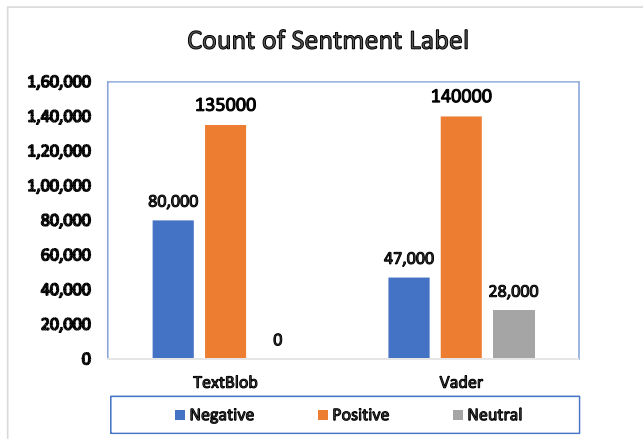


Fig. 5. Comparison of sentiment labelling using Textblob and Vader.

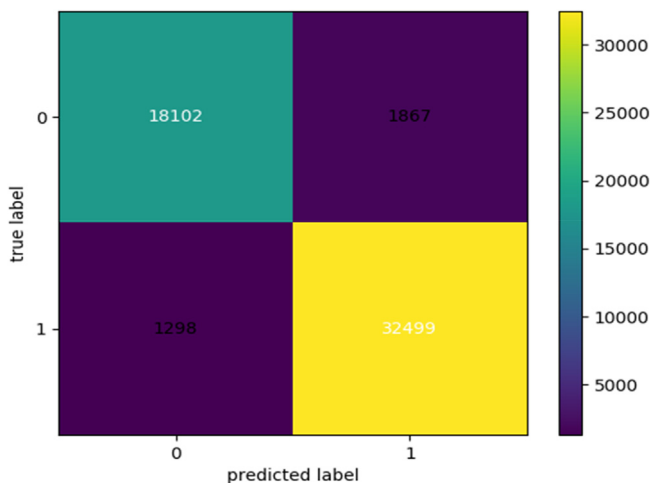


Fig. 6a. Prediction of sentiment labels: Confusion Matrix-Textblob.

5.1. Comparison of sentiment labelling

The aspect-based tokenization process generates the sentiment labels. Three typical review sentences selected from the data set

are highlighted in Table 2, from the drug review data set for easier understanding of computation. The highlighted words in the following review sentences are the aspect terms. To calculate the polarity scores with the aspect-based tokenisation per word is a tuple of polarity and subjectivity to be obtained. Then to be labelled as negative or positive based on the rule embedded in the code.

The consolidated report on the tokenization of dataset of size 215 k reviews is presented in the histogram as in Fig. 5. A basic statistical representation of histogram is drawn to compare both the methods quantitatively. In the figure the histogram column of Textblob count is observed as 'zero' in place of neutral label, because the set rules are for binary labels.

5.2. Prediction of sentiment labels with confusion matrix

The confusion matrix clearly differentiates the true positive and true negative values in both the methods, but the total number of values differ slightly because of the little subset of neutral values branched out from true negative and true positive values in the results of Vader method.

The measure of true positive and true negative for both methods are compared in Figs. 6a and 6b as confusion matrix.

5.3. Model category table

The experiment is modelled with the support of Logistic Regression, Naïve Bayes, SVC and Random Forest Classifier which are the standard classifier methods. Higher value of accuracy is found in Linear SVC (using rbf as kernel) and Logistic Regression in Textblob method. However, the accuracy values are found to be less in Vader method. Cross validation among different classification methods are computed as per number of folds which are tabulated in Table 3.

The Linear SVC and multinomialNB classifiers perform better in Vader method whereas the RFC and Logistic regression perform better in TextBlob method.

These symbolic supervised learning methods are compared and applied to human language technology, which often explains about higher values in the output of distinct algorithms.

The statistical representation of measuring the scores of classification models by depicting numerical data in group, the box plot accuracy for both the methods are in Fig. 7. The observed score is

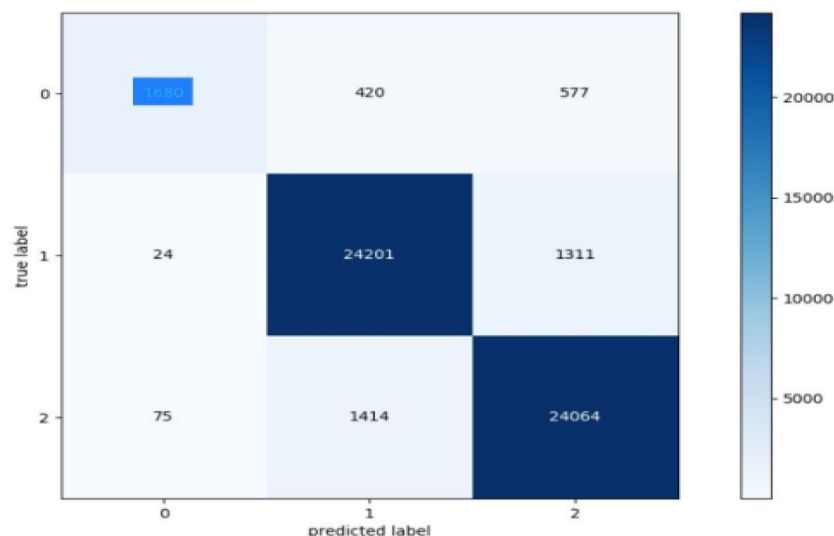


Fig. 6b. Prediction of sentiment labels: Confusion Matrix-Vader.

Table 3

Model category details.

Sl no.	Model Name	Fold_idx	Accuracy -Textblob	Accuracy Vader
1	RandomForestClassifier	0	0.625439	0.634026
2	RandomForestClassifier	1	0.625439	0.655035
3	RandomForestClassifier	2	0.625439	0.643596
4	RandomForestClassifier	3	0.625453	0.637427
5	RandomForestClassifier	4	0.625453	0.649671
6	LinearSVC	0	0.946505	0.932464
7	LinearSVC	1	0.944459	0.933532
8	LinearSVC	2	0.943389	0.935089
9	LinearSVC	3	0.943876	0.937622
10	LinearSVC	4	0.947875	0.935761
11	MultinomialNB	0	0.671030	0.81964
12	MultinomialNB	1	0.671890	0.818566
13	MultinomialNB	2	0.671053	0.82117
14	MultinomialNB	3	0.672347	0.822189
15	MultinomialNB	4	0.673510	0.822999
16	LogisticRegression	0	0.895590	0.888269
17	LogisticRegression	1	0.894148	0.889220
18	LogisticRegression	2	0.895171	0.890056
19	LogisticRegression	3	0.895378	0.891984
20	LogisticRegression	4	0.897029	0.891191

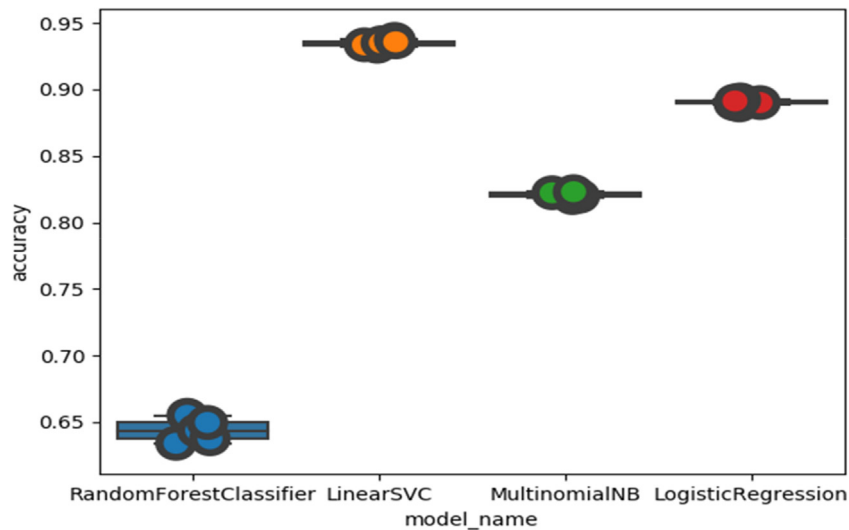
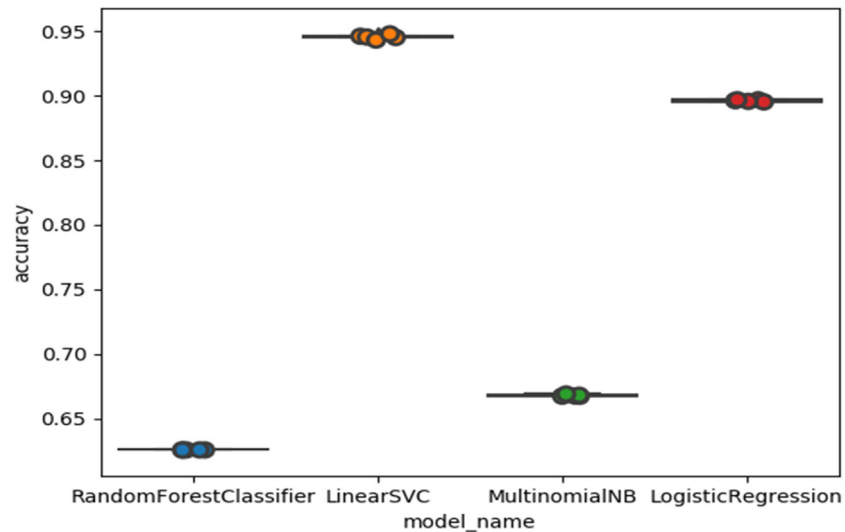
**Fig. 7.** Boxplot on Accuracy scores from both the methods Textblob Model, Vader Model.

Table 4

Summarised brief scores of both the methods.

Sl No	Model Name	Scores of Vader analysis	Score of Textblob analysis
1	LinearSVC	0.934894	0.945621
2	LogisticRegression	0.890144	0.896203
3	MultinomialNB	0.820913	0.667590
4	RandomForestClassifier	0.643951	0.625445

tabulated in Table 4. In Textblob method the accuracy of Linear SVC classification is on the higher side.

5.4. Classification report: accuracy

The classification report in Table 5, indicates that the positive precision by the text blob is little higher than the Vader method, and also the recall value is at higher performance than Vader method.

The observations are as follows

- Positive recall means 63% of positive labels are selected by Vader method but by Textblob it is selected as 96% of the times it should have been selected.
- Negative recall means negative labels are selected 94% but by Textblob it is 91% of the times it should have been selected, with weighted average of 93% in Vader method and in Textblob it is 94%.

- This model is quite good at classifying error, it seems to be weaker in Vader method of classifying positive class with recall percentage of 63% whereas it is 94% in text blob.

5.5. Count of sentiment labels by disease category

The visualization of the results in a graph is important where the intelligence in data is derived and it is easy to make definitive sentences in concluding the drug usage review. The visualization chart displayed in Fig. 8, clearly highlights the top constructive reviews as “Birth control”, “Depression” and “Anxiety”. The selected data source of machine-learning-databases indexed as 00462 (UCI) collected from the set as mentioned in Section 4 of this paper. The polarity score ranges from -1 to 1 , the subjectivity from 0.0 to 1.0 . This process is widely used in Textblob. But in Vader analysis it is termed as neutral, positive and negative, when it is $\geq (0.05)$, $\leq (-0.05)$ and rest, respectively.

6. Conclusion

The work aims at finding the polarity score in terms of subjectivity and polarity using aspect-based sentiment analysis through Textblob and Vader NLP packages. The dataset of drug review sentences from UCI repository is used. The Textblob methodology is being used to identify the aspects of review sentences and cate-

Table 5

Classification report: Accuracy by recall [Vader] and [Textblob].

Sl. No.	Parameters	Vader				Textblob			
		precision	recall	f1-score	support	precision	recall	f1-score	support
1	Negative	0.93	0.94	0.93	25,553	0.93	0.91	0.92	19,969
2	Positive	0.94	0.63	0.75	2677	0.95	0.96	0.95	33,797
3	Neutral	0.93	0.95	0.94	25,536	—	—	—	—
4	Accuracy	—	—	0.94	53,766	—	—	0.94	53,766
5	macro avg	0.93	0.84	0.88	53,766	0.94	0.93	0.94	53,766
6	weighted avg	0.93	0.93	0.93	53,766	0.94	0.94	0.94	53,766

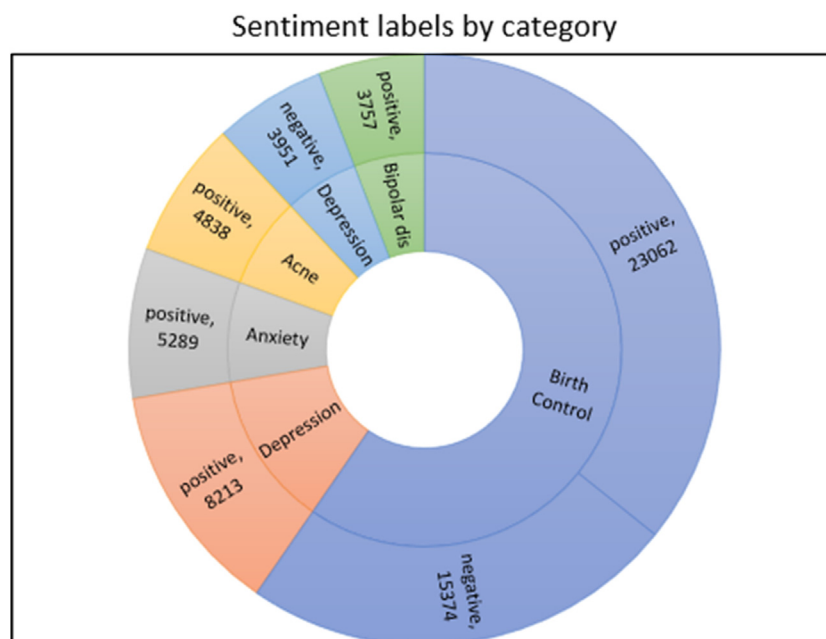


Fig. 8. Sunburst chart with count of sentiment labels with details by disease category (details).

gorised as negative and positive to decide upon the drug usage feedback from the users.

The accuracy scores of the algorithm displays higher values for Textblob method compared to Vader method. The classification is carried out by Linear Support Vector Classifier (SVC) using radial basis function as kernel.

With respect to the drug review data used in the analysis the top constructive reviews are from “Birth control”, “Depression” and “Anxiety” disease categories respectively. The polarity scores observed from Textblob are in the range of -1 to 1 , and from 0.0 to 1.0 for the subjectivity. However, in Vader analysis the polarity scores are observed as neutral, when the score is ≥ 0.05 , positive when the score is ≤ -0.05 , and it is negative for rest of the values respectively.

7. Future work

The pre-processing methodology can be enhanced for further analysis of sentiment behaviour of human language for the aspect of irony and sarcasm. The best fit classification model also can be tested in the identification of aspects in sentences. All the libraries used in sklearn and NLTK can be used with better confidence for arriving promising results. Future work can be extended by doing iterations with values of 0.0001 , and variations can be observed. A next level of NLP can be processed on the resultant dataset i.e. the review sentences of negative labelled to make any contraindication and to map relative priorities to map with the clinical trials. This enables ways to identify symptoms and target the root causes to highlight fine-tuned potential drug in pharmacy domain.

The extension of the work can also be carried out with deep learning methods by considering the condition specific information in the reviews. The Sentic-Prom methodology can be developed to highlight the benefits of specific healthcare domain features.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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